Deliverable 3 - COST ESTIMATE REPORT
EduROOM: Cloud-Based Classroom Reservation System

1. **Architecture Summary**

EduROOM is a cloud-based classroom reservation and scheduling solution deployed on Microsoft Azure. The system is structured around clearly defined public and private boundaries, where only the App Service is exposed to the internet while all backend services are isolated within a secured Virtual Network (VNet). This design ensures that external users can access the application safely without direct exposure to sensitive infrastructure components.

Within the backend layer, all core services are integrated through this secure network environment. The App Service acts as the central application runtime and connects to Azure Database for MySQL for persistent data storage, Azure Key Vault for secure management of secrets such as database credentials and API keys, and Azure Communication Services for handling email delivery, OTP verification, and notifications. The Virtual Network enforces network-level isolation, ensuring that private resources such as the database and Key Vault are not publicly accessible and can only be reached through controlled internal communication paths.

Security is further strengthened through Managed Identity, which allows the App Service to authenticate directly with Azure services without storing credentials in code or configuration files, eliminating hardcoded secrets and enforcing identity-based access control across all service interactions. For scalability and reliability, the application is deployed across three App Service instances with Microsoft Azure's built-in load balancer distributing traffic to ensure high availability and fault tolerance. The deployment process is fully automated using GitHub Actions CI/CD, enabling continuous integration and deployment whenever changes are pushed to the repository, which streamlines development and ensures consistent updates to the production environment.

2. **Itemized Cost Breakdown**

Azure Services	Configuration	Estimated Monthly Cost
App Service	Basic Tier B1	$13.14
SQL Database	Flexible Server Deployment, Burstable Tier B1MS	$19.64
Key Vault	10,000 operations	$0.03
Private Link	1 Endpoint with 100 GB inbound/outbound data	$9.30
Communication Services	10,000 emails per month	$2.50

Total Estimated Monthly Cost
Estimated Total: $44.61/month

Estimated Annual Cost
Estimated Annual Total: $535.26/year

3. Azure Pricing Calculator

EduROOM Estimate

App Service	Basic Tier; 1 B1 (1 Core(s), 1.75 GB RAM, 10 GB ...	Upfront: $0.00	Monthly: $13.14	Annual: $157.68
Azure Database for MySQL	Flexible Server Deployment, Burstable Tier, 1 B1...	Upfront: $0.00	Monthly: $19.64	Annual: $235.62
Key Vault	Vault: 10,000 operations, 0 advanced operation...	Upfront: $0.00	Monthly: $0.03	Annual: $0.36
Azure Private Link	1 Endpoints X 730 Hours, 100 GB Outbound dat...	Upfront: $0.00	Monthly: $9.30	Annual: $111.60
Azure Communication Services	Phone numbers leasing: 0 local and 0 toll-free ...	Upfront: $0.00	Monthly: $2.50	Annual: $30.00

Support

SUPPORT:

Basic (Included) $0.00

Select your program/offer

LICENSING PROGRAM:

Microsoft Customer Agreement (MCA)

Selected billing profile:
None selected (change)

Show Dev/Test Pricing

Estimated upfront cost	$0.00
Estimated monthly cost	$44.61
Estimated annual cost	$535.26

Figure 1. The Azure Pricing Calculator estimate for the Azure cloud deployment architecture of the EduROOM.

4. Cost Optimization Notes

In order to lower operating expenses while providing a satisfactory system performance, security and reliability, several strategies were deemed.

4.1 Azure for Students Subscription

The deployment utilizes the Azure for Students subscription, which provides $100 in free educational credits. This significantly reduces development and testing costs, covering the majority of expenses incurred during the project lifecycle.

4.2 Basic Tier App Service

The application is deployed on the Basic B1 App Service Plan due to the relatively small expected academic workload and the acceptance to deploy the application as a prototype.

4.3 Burstable MySQL Tier

Azure Database for MySQL uses Burstable B1MS pricing tier to lower the operational costs of your database without compromising database performance for low to moderate traffic.

4.4 Scale Down During Off-Hours

Since EduROOM is primarily used during school hours (8:00 AM – 5:00 PM, Monday to Friday), scaling the App Service down to 1 instance during off-hours and weekends could reduce compute costs during idle periods.

4.5 Free Tier Services

Several Azure services used in EduROOM offer free tier limits that cover typical student project usage. Azure Key Vault provides the first 10,000 operations per month at no cost, Azure Communication Service includes the first 100 emails per month for free, and Virtual Network inbound data transfer is free. By staying within these limits, the project eliminates costs for these three services entirely compared to paid usage.